

Renewable-Energy Investment Decision

CORE DECISION

Should Axion secure additional renewable electricity through further PPAs or invest directly in renewable-generation capacity?

The viability of additional solar investment should be assessed together with expected customer demand profiles because AI-focused customers generally require higher power densities and cooling requirements than traditional colocation customers. Renewable energy can supplement supply, but its intermittent nature means it may not fully satisfy the continuous power requirements of AI workloads without sufficient baseload and backup infrastructure.

Contents

1. Trigger key facts and important terms.....	2
The Decision.....	3
Initial Feasibility.....	3
Incremental cash flows + NPV.....	3
Principal assumptions and risk.....	6
Detailed assessment.....	8
Commercial and technical due diligence.....	8
Financing Terms and debt service capacity.....	9
Final Decision.....	10
Final Project Value and Shareholder Returns.....	10
Payback period (Discounted).....	10
Sensitivity analysis.....	10
Break-even analysis.....	11
Qualitative comparison.....	12
Wheeling and reliability.....	13
Potential APC question 1.....	14
Potential APC question 2.....	14
Potential APC question 3.....	15

1. Trigger key facts and important terms

Fact	Exam significance
Renewable electricity currently procured through (Power Purchase Agreements) PPAs as well as onsite renewable generation (<i>not the main source, supplementary</i>)	Axion already has practical exposure to the PPA model In addition to this, the company's existing onsite solar infrastructure provides management with practical experience regarding solar generation, maintenance requirements, energy production patterns, regulatory approvals and integration with existing power systems. Consequently, while expansion projects remain subject to construction, permitting and operating risks, management is likely better positioned to assess and manage these risks than an entity with no prior renewable energy experience.
45 MW installed IT capacity, 35 Utilized, 40 Contracted. <i>Don't know spare capacity as need to take failover headroom into account, as well as supported projections of future utilisation to calculate.</i>	We currently have limited capacity and are also making decisions about our customer mix. Customer mix should be considered when evaluating solar expansion, as AI-focused customers typically require significantly higher power densities and cooling capacity than traditional colocation customers.
Additional PPA option	Limited upfront capital; price, volume and supply terms are contractual; flexibility depends on the agreement.
Direct investment option	Requires upfront capital and funding; Axion participates in project risks and returns; more construction, operating and governance involvement.
Off-site generation and wheeling	Land/network constraints may require remote generation and grid-based allocation
Long-term offtake	A contract under which a future customer commits to buy the electricity that will be produced by the project once it is operational. Due to the risks explained, it is most likely that this occurs in combination with the direct investment option.
Reliability	Renewable generation does not replace grid connections, UPS/batteries or backup generation.
Installed IT capacity	The customer IT load Axion's completed facilities can support. Increasing it requires broader data-centre infrastructure, not merely more electricity.
Electricity supply	The sources and network arrangements that provide power to the facilities.
PPA	A long-term contract under which Axion buys electricity from an independent producer. Axion takes: - Electricity price risk (depending on contract structure) - Volume risk (Renewable generation may vary) - Counterparty risk (dependency risk that the other party to the contract—the renewable-energy supplier or IPP—cannot or does not fulfil its obligations to Axion. → If the supplier defaults or underperforms, Axion may have to purchase replacement electricity from the grid or another supplier, potentially at a higher price. This could increase costs and threaten the reliable electricity supply required by its data centers.) Axion does not take on (as these remain with the solar/wind developer) - Construction risk - Permitting risk - Technology risk - Operating and maintenance risk - Asset obsolescence risk
IPP	An independent power producer, not a separate decision method.
Wheeling	Grid-based commercial allocation/reconciliation of remotely generated electricity; electrons are not individually traced.

The Decision

The two alternatives to model, if facts permit

1. Alternative 1: enter into additional PPA arrangements.
2. Alternative 2: invest directly in renewable-generation capacity, alone or with a technical partner.

The decision should answer four questions

Question	Application
Value	Which option has the superior risk-adjusted economic outcome?
Risk allocation	Who bears development, construction, operating, generation, price, volume, curtailment and counterparty risk?
Strategic fit	Should Axion own generation assets or preserve focus on data-center infrastructure? Axion's core business is data centres; the renewable investment is an enabler. Renewable energy can only supplement, due to the nature of renewable energy and the power required for data centres.
Capital allocation	Is direct generation a better use of scarce capital and management capacity than further data-center expansion?

Initial Feasibility

If we are following their method, we would just calculate the incremental cash flows pretax and discount at a pretax discount rate. NB to see on the day **

Incremental cash flows + NPV

Additional PPA Forecast will be for contract period (no terminal value needs to be calculated as it is a fixed life - typically 3-5 years.)	Direct investment See if fixed life or not on the day, depends on details** Typically, fixed life = 25-30 years, depending on tech.
PPA tariff and year-on-year escalation (cost of PPA)	Initial generation CAPEX and grid-connection expenditure (CF0)
Volume driver = contracted volume (<i>What would multiply PPA tariff by...</i>)	Operating and maintenance costs (CF1 onwards) Working capital costs and will have the recovery at the end... just sum them up.
Wheeling, network and administration charges	Replacement CAPEX and insurance Replacement capital expenditure required to sustain the generation capacity of the renewable asset. Depreciation can be used as a reasonableness check when assessing whether sufficient reinvestment has been included in the model. Should be at least in line with depreciation → although accounting measure shows managements expectation of the life of the asset.
Transmission/distribution losses where relevant <i>** We take this into account through the volume as seen below</i>	Wheeling/network costs for remote generation <i>Wheeling will be applicable as solar plant will not be onsite - see related costs!</i>

Additional PPA Forecast will be for contract period (no terminal value needs to be calculated as it is a fixed life - typically 3-5 years.)	Direct investment See if fixed life or not on the day, depends on details** Typically, fixed life = 25-30 years, depending on tech.
Cost/credit for unused or excess electricity <i>Where sale of excess cannot occur, the cost of excess is already indirectly taken into account. Would have the cost based on contracted volume, your PPA tariff and then the revenue from contracts entered into.</i> <i>However in some instances, excess electricity can be sold (Wheeling elsewhere, selling into the grid or receiving a credit for excess) → the expected proceeds or credits from surplus electricity should also be incorporated into the valuation.</i> To determine the spare capacity: <i>Expected generation (which actually arrives) less expected demand less reserve or buffer requirements = potential surplus</i> <i>In assessing potential surplus electricity and associated credits, management should consider forecast customer demand over the life of the PPA rather than current utilisation levels alone. The analysis should incorporate expected growth in contracted capacity, future expansion phases, customer demand profiles, reserve requirements and redundancy obligations, as these factors determine the amount of electricity that will ultimately be available for resale or generate excess-capacity credits.</i> ** whether you can sell is governed by PPA arrangement.	Proceeds from surplus electricity only if supported by facts - i.e. enter into agreements to sell surplus electricity under PPAs ← Same idea

**** transmission and distribution losses = physical electricity losses that occur when electricity travels through the network**

$$\begin{array}{l}
 \text{Required demand} \\
 \div \\
 (1 - \text{loss \%}) \\
 = \\
 \text{Energy to be purchased}
 \end{array}$$

If they give us the demand that we need to service and from there we need to determine energy to be purchased to plug into valuation - remember to take transmission losses into account!

PPA Contracted Volume additional thinking **how the calc can change depending on contract terms:

Scenario 1: Delivery measured at your connection point (common approach) Contract will say "Axion pays for electricity delivered at the delivery point"	Scenario 2: Delivery measured at the generator Contract will say "Axion purchases all electricity generated"
Suppose: - Solar farm generates 100 MWh 5 MWh lost in transmission 95 MWh arrives at Axion Axion pays for 95 MWh, not 100 MWh	Suppose: - Solar farm generates and we contract for 100 MWh 5 MWh lost in transmission 95 MWh arrives at Axion Axion pays for 100 MWh, not the 95
What this means for NPV: When calculating NPV, the relevant cost will be the 95* cost per unit.	What this means for NPV: When calculating the NPV, the relevant cost would be the 100* cost per unit

Tax calculation ** link to Tatum specific - bring in once read through

PPA Option	Direct Investment Option
Included in taxable income	Included in taxable income
Electricity savings / revenue benefit	Energy benefit
Less PPA payments	Less O&M
Less wheeling / admin costs	Less tax allowances
Add surplus electricity sales	Add recoupment (if applicable)
	Add taxable capital gain (if applicable)
Excluded from tax calculation	Excluded from tax calculation
Working capital increase	CAPEX
Working capital recovery	Working capital increase
	Working capital recovery
	Disposal proceeds (not directly)
Taxable income	Taxable income

Income tax to include in the net present value calculation

Compare PPAs with direct generation investment using 27% income tax and 21.6% capital gains tax. Apply adjustments only to after-tax cash flows discounted at a post-tax WACC.

Cash flow line	Additional power purchase agreements	Direct investment in generation capacity
Electricity benefit/ saving	Included in taxable income	Included in taxable income
Payments for electricity under the agreement	Deductible in full under section 11(a)	Not applicable
Wheeling, network and administration charges	Deductible in full under section 11(a)	Deductible in full under section 11(a) where generation is off site
Operating, maintenance and insurance costs	Not applicable	Deductible in full under section 11(a)
Proceeds from surplus electricity sold, wheeled or credited	Included in taxable income when it accrues	Included in taxable income when it accrues
Initial capital expenditure	Not applicable	Not deductible
Capital allowances	None. No asset is owned, and sections 12B, 12BA and 11(e) each require ownership	Section 12B at 50/ 30 / 20 per cent over three years, or 100 % in year one for photovoltaic solar not exceeding 1 megawatt. Section 12U at 100 % on new roads and fences. Section 13quin at 5 % on new buildings. Section 11(e) on the grid connection and transformers
Land	Not applicable	No allowance is available under any section
Amounts paid in advance under the contract	Section 23H spreads the deduction over the period to which the expenditure relates	Section 23H spreads the deduction over the period to which the expenditure relates
Interest on funding	Not applicable	Deductible under section 24J as it is incurred. Section 23M limits the deduction where the lender is in a controlling relationship and is not subject to South African tax on the interest
Replacement capital expenditure	Not applicable	A qualifying replacement asset starts a fresh section 12B allowance. The asset replaced gives rise to a recoupment under section 8(4)(a) or an allowance under section 11(o)

Movement in and recovery of working capital	No tax effect	No tax effect
Disposal at the end of a finite life	Not applicable	The proceeds are not themselves taxable. The recoupment under section 8(4)(a) is taxed at 27%, and any excess of proceeds over cost is a capital gain taxed at 21.6%

The main tax advantage of direct investment is the capital allowance. Axion can use the full tax saving in years 1-3 only if it has enough taxable income; otherwise, some of the saving is delayed.

Under section 20(1)(a) a balance of assessed loss brought forward may be set off only against the higher of R1 million or 80 per cent of taxable income, so allowances that create an assessed loss are deferred.

Exclude or treat carefully

- **Sunk costs** are already incurred → exclude them. **Example:** exclude R2 million already paid by Axion for a feasibility study or legal advice.
- **Common costs** incurred under both options → exclude them. **Example:** unchanged grid, UPS or backup-generator costs do not affect the comparison.
- **Allocated overheads** that do not change → exclude them. **Example:** exclude existing head-office salary allocations, but include the salary of a new project manager required for direct investment.
- Financing cash flows → exclude loan receipts, interest and repayments from a project NPV using WACC to avoid double counting. **Example:** include Axion's solar-plant CAPEX, but exclude the related R300 million loan flows.
- **Non-cash accounting entries** → exclude them unless they affect tax or another cash flow. **Example:** exclude depreciation, but include the related tax saving.

Calculation of NPV

- Calculate sum of each year
- Calculate your project specific WACC rate (make sure rate use is however aligned to whether we are doing pretax cash flows or not)
- If it doesn't have a finite life, must calculate a **terminal value**.

Terminal value

Terminal value represents the value of all future cash flows beyond the explicit forecast period.

Terminal value should be based on sustainable, recurring cash flows and assumptions that are expected to continue indefinitely or for the remaining economic life of the asset.

One-off items must be excluded from terminal value calculations, for example:

- Working capital recoveries (as there is none, as project goes on forever)
- Disposal proceeds (as there is none, as project goes on forever)
- Terminal CAPEX replacements (not going to happen in each period, which is what will happen if plot in terminal)
- Restructuring costs (once off... not sustainable)
- Once-off tax benefits or liabilities (not sustainable)
- Once-off surplus electricity sales (not sustainable)

Calculation:

Perpetuity growth model:

$$TV = \frac{FCF_{n+1}}{WACC - g}$$

Or terminal value could be based on exit multiple, EBITDA typically used in data infrastructure

Source	EV/EBITDA
Renewable-energy sector benchmarks	~14.5x median dealmatrix
Renewable-energy deal discussion	~11.1x median Q4 2024 phoenixstrategy
Public renewable companies	often 9x-18x+ depending on growth and risk phoenixstrategy

Principal assumptions and risk

PPA

Assumption	Risk if assumption is incorrect	Impact
Forecast customer growth, customer mix, pricing assumptions and historic utilisation trends provide a reliable basis for determining future electricity requirements. The revenue forecast appropriately reflects expected changes in customer demand, including the proportion of AI-focused customers (revenue reflects the expected change in customer mix and resulting pricing considerations.) In turn, the contracted PPA volumes have been determined using utilisation forecasts, expected customer demand, appropriate redundancy capacity and buffer requirements, and consistent with the revenue profile assumed in the valuation. <i>We already have customer contracts, so we are not completely in the dark regarding demand. Some growth is forecast and therefore utilisation must be projected. Customer mix is projected + Pricing assumptions in customer contracts are projected.</i> <i>AI customers have higher rack density and power requirements. The revenue model assumes a certain customer profile.</i> <i>The electricity procurement model must reflect that same customer profile.</i> <i>The PPA volume is based on expected utilisation plus redundancy/headroom requirements. The cost model and revenue model must therefore tell the same story.</i>	Future customer requirements and power-density profiles differ from assumptions, particularly where AI-related demand grows faster or slower than expected.	Axion may contract for insufficient or excess electricity capacity, resulting in take-or-pay costs, underutilised contracted volumes, increased purchases of supplementary electricity, higher operating costs, or an inability to support future customer growth. In addition, the electricity procurement assumptions may become inconsistent with the customer demand and revenue assumptions underpinning the valuation.
Actual electricity delivered under the PPA is consistent with forecast generation after considering curtailment, wheeling losses and transmission constraints.	Curtailment, wheeling losses, grid constraints or supplier underperformance	Renewable energy benefits are lower than expected and reliance on Eskom/grid electricity

	reduce delivered electricity.	increases + leads to higher costs.
PPA tariffs , wheeling charges and other network costs remain within forecast levels.	Electricity tariffs, wheeling charges or regulatory costs increase above forecast.	Operating costs increase and project returns decline.
Surplus electricity can be sold , wheeled, transferred or credited where permitted by the PPA.	Contractual restrictions prevent monetising excess electricity.	Axon incurs take-or-pay costs on unused electricity volumes.
The supplier remains financially and operationally capable of meeting contractual obligations.	Supplier financial distress or operational failures disrupt supply. (counterparty risk)	Additional power must be sourced elsewhere , potentially at a higher cost .

Investment in plant

Assumption	Risk if assumption is incorrect	Impact
Forecast customer growth, customer mix, utilisation patterns and AI workload requirements provide a reliable basis for estimating future electricity demand and the economic benefits of the renewable investment. The projected demand profile is consistent with the revenue forecast and expected power-density requirements of customers.	Actual customer demand, utilisation trends or customer mix differ from forecast , particularly if AI-related demand develops differently from expectations.	Electricity generated may be insufficient or excessive relative to actual requirements, reducing expected savings, resulting in underutilized generation capacity or requiring additional electricity purchases from the grid
Operating, maintenance and replacement costs remain within forecast levels.	Higher O&M costs or earlier -than-expected equipment replacement requirements.	Project returns are lower than forecast and the investment may not achieve the expected payback period or NPV.
Generation output will achieve forecast performance and output levels.	Weather conditions, equipment failures, curtailment or transmission loss or operating inefficiencies reduce generation	Additional electricity must be purchased from the grid or under PPAs, reducing expected savings and project returns.
Construction will be completed on time and within budget.	Delays, contractor issues or cost overruns occur.	Capital costs increase and expected benefits are deferred.
Required permits , land rights, grid access arrangements and regulatory approvals remain available and are obtained within the expected timeframe.	Regulatory changes, approval delays or grid access constraints occur.	Project implementation may be delayed, costs may increase and the economic viability of the investment may deteriorate.

Both AI risks and assumptions

assumption	Risk	Impact
AI, cloud and colocation growth will occur in line with forecast and power can be scaled as required.	AI customers require significantly higher rack densities and power consumption than anticipated.	Electricity demand may outpace contracted or self-generated renewable capacity, requiring expensive emergency capacity procurement and accelerating future expansion spend.

Detailed assessment

Validate the complete investment case

Commercial and technical due diligence

Commercial due diligence assesses whether the investment is worth doing, while technical due diligence determines whether it will work as expected.

→ Commercial due diligence assesses whether the investment makes financial and business sense by reviewing factors such as pricing, contracts, demand, funding and expected returns.

→ Technical due diligence assesses whether the project can be constructed, operated and generate the expected electricity output.

PPA

Due Diligence Area	What Axion should assess	Why it matters
Commercial	Review PPA tariff structure, escalation clauses, wheeling charges, take-or-pay obligations and contract term.	Determines whether pricing remains economical over the contract term → high escalation or charges could increase costs and reduce the PPA's NPV.
Commercial	Assess forecast electricity demand against contracted volumes.	Reduces the risk of paying for unused electricity → excess contracted volumes may create take-or-pay costs.
Commercial	Evaluate ability to sell, wheel, transfer or bank surplus electricity.	Excess electricity has value only if contractual rights exist → without resale, wheeling, transfer or banking rights, Axion may receive no benefit from surplus power.
Commercial	Assess supplier financial stability and track record.	Reduces supplier-default and disruption risk → failure may force Axion to buy replacement electricity at a higher cost.
Commercial	Review contractual performance guarantees and remedies.	Determines Axion's protection if electricity is not delivered → guarantees, penalties or damages may recover losses and encourage supplier performance.
Technical	Evaluate expected generation output of the facility.	Confirms whether forecast electricity volumes are achievable → lower output reduces renewable benefits and increases grid purchases.
Technical	Assess transmission constraints, wheeling arrangements and expected losses.	Actual electricity received may be lower than generated → losses or network constraints reduce usable MWh and may increase the effective cost per unit.
Technical	Assess historical curtailment levels and grid reliability.	Curtailment may reduce delivered electricity → Axion may need additional grid power, lowering savings and the PPA's value.
Technical	Confirm that generation profiles align with Axion's demand profile.	Data centres require continuous electricity supply → mismatched renewable output means Axion must retain grid, UPS and backup capacity.

Direct investment

Due Diligence	What Axion should assess	Why it matters
Commercial	Assess forecast customer demand,	Determines whether sufficient demand exists to justify the investment → weak demand may leave

I	utilisation growth and AI-driven power requirements.	Axion with underutilized generation capacity.
Commercial	Evaluate project NPV, IRR, payback period and funding requirements.	Confirms whether shareholder value is created → a negative NPV or inadequate IRR would indicate that returns do not justify the investment risk.
Commercial	Assess available funding sources and impact on liquidity.	Large renewable projects require substantial capital → funding may strain Axion's liquidity or restrict data-center expansion.
Commercial	Assess expected future electricity prices.	Determines potential savings compared with external electricity purchases → lower/higher future tariffs may reduce/raise the value of owning the plant.
Commercial	Evaluate opportunities to monetise excess generation.	Additional revenue may improve project economics → Hwr, without customers or contractual rights, surplus electricity may generate no value.
Technical	Assess expected generation output and resource availability (solar irradiation, wind conditions, etc.).	Determines achievable energy production → lower resource availability reduces savings and increases grid purchases.
Technical	Review engineering design and generation capacity.	Confirms the asset is appropriately sized for future demand → oversizing wastes capital, while undersizing requires additional electricity purchases.
Technical	Assess construction plans, timelines and contractor capability.	Delays and cost overruns reduce returns → savings begin later and additional CAPEX lowers the project's NPV.
Technical	Evaluate O&M requirements and lifecycle costs.	Significant maintenance costs affect investment returns → higher lifecycle cash outflows reduce NPV and may require earlier replacement spending.
Technical	Assess asset performance, degradation and replacement requirements.	Generation assets lose efficiency over time → declining output reduces savings and may trigger replacement CAPEX.
Technical	Confirm grid connection requirements and potential curtailment exposure.	Connectivity constraints may reduce realised benefits → curtailment or delayed grid access can lower usable output and postpone savings.

Financing Terms and debt service capacity

There is generally no direct equivalent to debt-service capacity for equity, as equity does not require mandatory repayments in the same way as debt. Equity is therefore more flexible and does not expose the company to default risk arising from fixed payment obligations. However, where shareholders have an established expectation of dividends or where the company has a history of regular dividend distributions, management should assess the impact of the investment on future distributable cash flows and dividend-paying capacity. In this respect, the assessment becomes similar to debt-service capacity, although the focus is on dividend sustainability rather than contractual repayments. The framework already addresses many equity-related considerations indirectly through the NPV analysis. A positive NPV indicates that the project is expected to create shareholder value and generate returns in excess of the required return on capital.

Additional equity-focused questions include → *something the current strict framework is missing*

- Whether existing shareholders have the financial capacity and willingness to contribute additional equity capital.
- Whether external equity funding or a strategic investor will be required.
- The extent of any shareholder dilution arising from a new equity issue.
- Whether the expected equity returns adequately compensate shareholders for the additional construction, operating and regulatory risks assumed under the direct investment option.
- The impact of the investment on future dividend-paying capacity and financial flexibility.

PPA - Financing terms

Financing Term	What Axion should assess
----------------	--------------------------

Consideration	
PPA tariff structure	Fixed price vs escalating tariff.
Contract term	Whether the contract term aligns with expected customer demand.
Take-or-pay provisions	Whether Axion must pay regardless of actual consumption.
Indexation/escalation	Inflation-linked increases and future affordability.
Security requirements	Deposits, guarantees or credit support required by the supplier.
Termination clauses	Cost of exiting the arrangement if demand changes.

PPA - Debt service capacity

Question	Why it matters
Can operating cash flows comfortably support future PPA payments?	PPA commitments become a fixed operating cost.
Can Axion absorb take-or-pay costs if utilisation falls below forecast?	Excess contracted volumes may still need to be paid for.
Can Axion absorb tariff escalations and wheeling charge increases?	Future electricity costs may exceed forecast.

Direct Investment - Financing terms

Consideration	What Axion should assess
Debt versus equity funding	Determine the most appropriate funding mix.
Interest rate	Fixed versus floating borrowing costs.
Loan tenor	Whether repayment periods align with project cash generation.
Covenants	Potential restrictions imposed by lenders.
Debt service coverage requirements	Ability to meet lender requirements throughout the project life.
Security requirements	Assets or guarantees required by lenders.

Direct Investment - Debt service capacity

Question	Why it matters
Will project cash flows cover principal and interest repayments?	Determines ability to service debt.
What happens if generation is lower than forecast?	Lower savings reduce cash available for repayments.
What happens if customer demand is lower than forecast?	Less electricity consumed means lower economic benefit.
Can Axion continue funding data centre expansion while servicing renewable-energy debt?	Competing capital requirements may create pressure on cash flows.
Will additional debt affect gearing levels and borrowing capacity?	Could limit future financing flexibility.

Final Decision

Determine whether the investment should proceed

Final Project Value and Shareholder Returns

Metric	Why
--------	-----

NPV (with tax)	Value created for shareholders
IRR	Percentage return generated
Payback period	How long before investment is recovered
Sensitivity analysis	Impact of downside scenarios
Scenario analysis	Best case / base case / worst case

Payback period (Discounted)

Year	PV cash flow	Cumulative
0	(100.0)	(100.0)
1	27.3	(72.7)
2	24.8	(47.9)
3	22.5	(25.4)
4	20.5	(4.9)
5	18.6	13.7

Sensitivity analysis

1. Base case – calculate NPV

Variable	Base Assumption
Electricity demand	100%
Generation output	100%
CAPEX	R500m
Discount rate	12%
NPV	R150m

2. Change **one** variable and calculate new NPV and calculate the **% change**. The change can result in potential upside of that % but also downside

Variable	Why it matters
Electricity demand / data-centre utilisation	Lower utilisation can create unused PPA commitments or underutilised owned generation.
PPA tariff and escalation	Changes in lifetime contractual electricity cost.
Grid tariff escalation	Changes avoided-cost benefits.
Generation output	Lower output reduces usable renewable electricity and avoids cost.
Direct-project CAPEX	Cost overruns reduce NPV and strain funding.
Wheeling charges , losses and curtailment	Reduce delivered economic benefit.
Operating & Maintenance and replacement costs	Increase lifetime ownership cost.
Completion timing	Delays the start of savings and can raise construction/financing exposure.
Discount rate	Tests the effect of project risk and long-dated benefits.

Additional Points:

- The different variables should be considered when performing a sensitivity analysis. The purpose of the sensitivity analysis is to perform a calculation for “what if” scenarios and see how these scenarios could impact the different investment options.

Break-even analysis

- Maximum acceptable PPA tariff or escalation rate.
- Maximum direct investment CAPEX.
- Minimum electricity demand or data-centre utilisation.
- Minimum annual usable generation.
- Critical grid-tariff escalation at which the preferred alternative changes.
- The break-even point is indicated by an NPV of 0
- In order to calculate the break-even point, start by calculating the Net Present Cost (NPC)
- This will then be used as the PV to calculate the annual PMT (income from the project in order to break even) see below.

Example:

NPC = -20mil

Use this to calculate the PMT amount:

PV = -20m

I = WACC

N = duration of project

FV= Residual value of project

PMT = ? = amount of income needed in order to break even

Liquidity and covenant impact [ADDED]

A positive NPV does not mean Axion can fund the project. Page 8 asks this as a separate question for a reason: the pipeline is R3 to R5 billion over five years, and borrowings have risen every year.

Measure	FY2024	FY2025	FY2026	What it shows
Interest cover (EBITDA / finance costs)	3.31	3.12	2.98	Falling every year despite EBITDA growth
Interest cover (operating profit / finance costs)	2.04	1.95	1.87	Below 2 times on the more conservative measure lenders use
Borrowings / EBITDA	4.23	4.36	4.35	At the level at which most lenders set the covenant
Borrowings / equity	0.78	0.85	0.90	Rising
Gearing, borrowings / (borrowings + equity)	43.8%	45.9%	47.3%	Rising
Capital expenditure / EBITDA	—	2.0	1.8	Capital spend is running at roughly twice operating cash generation

What to test on the day

- Headroom: recompute each ratio including the new debt and the new EBITDA, in the worst year of the forecast, not on an average.
- Debt service cover: project cash flow available for debt service divided by principal plus interest, tested each year, not once.
- Timing mismatch: debt is drawn and interest starts during construction, while the electricity benefit only starts at commissioning.
- Consequences of a breach: cure rights, an equity cure, and whether a breach on a project facility cross-defaults to the existing group facilities.
- Opportunity cost: every rand of debt capacity used on generation is not available for the data centre capacity expansion, which is the GROW priority.

Applied to Axion: the group is already at about 4.35 times borrowings to EBITDA and below 2 times operating profit interest cover, before any renewable energy spend. Debt-funding a plant of this size pushes both measures the wrong way during construction, when there is still no electricity benefit. That is the strongest financial argument for the PPA route, or for a structure in which the debt genuinely sits outside the group. It should be stated in any recommendation.

Risk allocation and safeguards [ADDED]

The risks are already identified in the assumptions tables above. Page 8 asks the second question: who carries each risk, and what safeguard reduces the residual exposure. State the safeguard, not only the risk.

Risk	Who should carry it	Safeguard to negotiate	Residual exposure to Axion
Construction delay and cost overrun	The contractor	Fixed-price turnkey contract, delay damages, a commercial operation date with remedies	Damages rarely cover a full loss of benefit
Generation below forecast	The producer under a PPA; Axion if it owns the plant	Performance guarantee, availability warranty, minimum yield undertaking	Replacement power still bought from the grid at a higher tariff
Curtailment and grid unavailability	Shared	Deemed generation clause, clear allocation in the wheeling agreement	Network charges may still be payable on power not received
Tariff escalation	Shared	Cap on escalation, or a fixed tariff for an agreed period	Escalation compounds over a long contract
Producer or contractor failure	The counterparty	Parent guarantee, step-in rights, security over the plant	Cost and delay of replacing supply
Demand below forecast	Axion	Volume flexibility band, the right to on-sell surplus, a shorter initial term	Take-or-pay cost on volume not used
Technology performance over the asset life	Axion if it owns the plant	Equipment warranties and an operating and maintenance contract with performance standards	A 25 year asset in a technology that continues to move

Required approvals [ADDED]

Approval	Why it is needed	What to confirm
Board approval	The delegation of authority framework sets the value at which board approval is required	Whether the project exceeds the threshold, and whether the board has approved the funding as well as the project
Shareholder approval	The memorandum of incorporation and any shareholders agreement. Axion has five shareholders and none holds control	Whether capital expenditure of this size, or entering a joint venture, is a reserved matter requiring shareholder consent
Lender consent	Existing facility agreements commonly restrict additional borrowing, the granting of security and disposals	Whether the new funding or security breaches a negative pledge and needs a waiver before signature
Companies Act section 45	Financial assistance to a related or inter-related company, which includes a subsidiary or a joint venture project company. Guarantees and loans to a project company fall within this	A special resolution passed within the previous two years, the board applying the solvency and liquidity test, and written notice to shareholders and any trade union. Non-compliance makes the decision void and exposes directors personally under section 77(3)(e)(iv)
NERSA registration or licensing	Generation above the registration threshold, and any wheeling arrangement	The timeline, because regulatory approval sits on the critical path to the commercial operation date and therefore to the first year of benefit in the model

Qualitative comparison

Something the current strict framework is missing

Consideration	Additional PPA	Direct Investment
Strategic fit	Supports Axion's objective of capturing growing customer demand while maintaining financial resilience. The option secures additional renewable-energy supply without requiring substantial capital investment, allowing management to focus resources on accelerating data-centre expansion and increasing available capacity. This is	Direct investment may align with Axion's long-term strategy if management believes access to electricity will become a critical enabler of future growth. As customer demand, AI workloads and power-density requirements increase, ownership of generation assets may provide greater influence over long-term electricity costs and

	consistent with management's objective of growing quickly without getting ahead of itself, as the arrangement may assist in meeting growing power requirements while limiting construction, operational and financing risks. However, Axion may have less direct control over a strategic input if electricity availability becomes a significant constraint on future growth.	future generation capacity. This could support Axion's ability to expand capacity and meet growing customer demand over the long term. The option is also consistent with management's view that decisions made today may shape Axion's position over the next twenty years. However, the investment requires significant capital, introduces additional operational complexity and may divert resources from core data-centre expansion initiatives.
Energy security	Improves long-term power security through contracted renewable supply but remains dependent on supplier performance and contractual protections.	Greater influence over generation capacity, maintenance decisions and future expansion, potentially improving long-term energy security.
Management focus	Allows management to focus on customer acquisition, data-center utilization, AI growth and capacity expansion.	Requires management attention to generation projects, technical oversight, contractors, maintenance and regulatory compliance.
Upfront capital	Typically requires limited upfront capital, preserving cash and debt capacity.	Requires significant upfront CAPEX and potentially additional debt funding.
Construction risk	Construction risk largely sits with the IPP, subject to contractual protections and performance guarantees.	Axion directly bears risks relating to delays, contractor performance, permitting issues and cost overruns.
Operating risk	Generation asset operations remain the responsibility of the IPP .	Axion bears or oversees operational performance, maintenance costs, degradation and component replacement.
Power-demand uncertainty	Easier to scale contracted volumes as future customer demand evolves.	Risk of investing in generation capacity that ultimately exceeds operational requirements or becomes insufficient if AI-driven demand grows faster than expected.
AI growth exposure	Additional electricity can potentially be contracted as AI workloads increase.	Incorrect forecasting of future AI-related power density may result in over- or under-investment in generation infrastructure.
Control	Relies primarily on contractual rights and supplier performance.	Greater control over design, expansion timing, maintenance and operational decisions.
Economic upside	Economic benefits are shared with the IPP through contract pricing.	Axion retains the benefit of lower generation costs if the project performs better than expected.
Price certainty	Provides visibility over future electricity costs through contracted pricing and escalation mechanisms.	Reduces exposure to supplier margins but exposes Axion to CAPEX, maintenance and performance risks.
Flexibility	Contract terms may create take-or-pay obligations, minimum-volume commitments and termination costs.	Long-term commitment to generation assets with potential stranded-capacity risk if demand forecasts change.

Wheeling and reliability

Simple explanation

Wheeling allows electricity generated at a remote renewable project to be commercially allocated to Axion through an existing transmission/distribution network (i.e. the electricity grid). Generator output and Axion consumption are measured and reconciled; individual electrons which have been produced at an off-site plant are not traced.

Essentially the concept of wheeling is feeding renewable energy into the grid off-site in order to use energy from the grid on-site.

Key Concepts in relation to Wheeling

- Process flow of wheeling: Renewable plant off-site -> feeds power into the grid -> Axion then receives power from the grid which it consumes
- Wheeling has costs. Eskom charges IPPs (private companies that generate electricity) to feed their power into the grid
- It is also important to take into consideration that the grid must have the capacity to allow the feed of renewable energy. Therefore, if there is a lack of capacity, there may be some power which has been generated which the IPP cannot feed into the grid.
- Metering is essential to track that Axion actually receives the power from the wheeling arrangement. There will be a meter at the off-site power plant and a meter at the data center. The 2 meters are reconciled to determine the power from the wheeling arrangement vs power from the grid (balancing generation against consumption).

Key consequences for Axion

Issue	Exam-day implication
Load shedding	A wheeling arrangement does not itself exempt Axion from load shedding; the data centre remains physically dependent on its connection and resilience infrastructure.
Metering	Reconcile generator exports, Axion consumption, time-of-use periods, losses, invoices and credits.
Losses	Generated MWh need not equal credited/usable MWh.
Residual supply	Renewable output may not match a continuous data-centre load, so grid/other supply remains necessary.
Curtailment	Grid constraints can reduce deliverable generation and change project economics.
Network capacity	Wheeling changes procurement/allocation; it does not by itself increase Axion's installed IT capacity.
Resilience	Grid, UPS/batteries and backup generation remain required under either remote-generation option.

Red flags requiring specific attention

- PPA starts materially before the related data-centre load ramps up.
- Headline tariff excludes material network, loss or residual-supply costs.
- Direct-generation case assumes 100% usable output or no curtailment.
- Base case depends on aggressive grid tariff escalation or perfect project completion.
- Termination amount or guarantee creates material contingent exposure.
- Direct investment is recommended without assessing capital opportunity cost.

Funding should reflect the project [ADDED]

Page 8 states that the funding structure appropriate for Axion's existing operations may not be appropriate for every new project. The three headings below are taken directly from page 8.

Project cash flows — predictability, duration, contractual support

Feature	Additional PPAs	Direct investment	Effect on debt capacity
Predictability	No new revenue stream. The effect is on Axion's operating cost	Generation varies with the resource, and the value of the saving varies with the grid tariff	Less predictable benefit supports less debt
Duration	Typically 3 to 5 years	25 to 30 years	Longer duration supports longer tenor debt
Contractual support	The PPA is itself the contract, but it is a purchase obligation, not a receivable	Only where output is sold under an offtake contract. A plant built for Axion's own consumption produces a cost saving, not contracted revenue	This is the decisive point. A self-consumption plant has no contracted revenue stream to pledge, so it supports materially less project debt than an independent producer selling under a long-term PPA

Target capital structure — project-specific mix

- The mix should reflect the risk of the project, not Axion's current group gearing of about 47%.
- A contracted, long-duration generation project can usually carry more debt than a data centre build. A self-consumption plant with no contracted offtake carries less.
- Whatever mix is chosen must then be used to regear the beta in the next section. Deriving the discount rate from one capital structure and funding the project at another is internally inconsistent, and is an easy point to be caught on.

Security and support — and why project-level funding does not transfer risk

Page 8 warns that project-level funding does not necessarily transfer risk away from Axion. Test that warning against what is actually signed.

- Sponsor support brings the risk back: completion guarantees, cost overrun undertakings and equity contribution agreements are all Axion obligations.
- Security typically includes a pledge of the shares in the project company, cession of the project contracts and a debt service reserve account funded by Axion.
- Project covenants trap cash: a distribution lock-up until a debt service cover ratio is met withholds the very cash flows Axion was relying on.
- Accounting follows substance. If Axion controls the project company it consolidates under IFRS 10 and the debt appears on the group statement of financial position. Joint control gives equity accounting under IAS 28, which keeps the debt off the face of the statements but not out of the group's risk. Axion already equity accounts its 40% interest in the Metroloop joint venture, so the precedent exists.
- Operational recourse remains regardless of structure. If the plant fails, Axion must still power the data centres and meet the 99.999% availability commitment.

Critique of the approach set out in Document 4 [ADDED]

Observation	Why it matters
The framework moves correctly from economic viability, to funding, to approval, and separates whether the project is worth doing from whether Axion can fund it.	This sequence is sound and should be acknowledged before criticising anything.
Initial feasibility uses pre-tax cash flows and a pre-tax discount rate.	Internally consistent, but it makes the capital allowance benefit invisible at the screening stage, which is exactly where direct investment differs most from a PPA. A project can screen out on a pre-tax basis and be positive after tax.
Project-specific cost of capital sits at initial feasibility, but the beta flow at the foot of page 8 requires the project's capital structure,	The process is iterative in practice and the pack presents it as linear. Say so, and state the assumed structure used at screening.

which is only settled later.	
Risk is identified at initial feasibility but allocated only at final decision.	Safeguards cost money. Price them into the cash flows rather than adding them as conditions after the NPV is calculated.
Digital tools are to be independently validated before use.	Any model or output presented on the day should be tested rather than accepted. This is stated on page 8 and is a likely task hook.
Nothing in the framework addresses capital rationing.	With a R3 to R5 billion pipeline and constrained funding, the question is not whether a project has a positive NPV but whether it is the best use of the next rand. Ranking by profitability index, not by NPV, is missing.

Project-specific cost of capital [ADDED]

The purpose of the three steps on page 8 is to obtain a discount rate reflecting the risk of the renewable energy project, not the risk of Axion's existing data centre operations. Axion is unlisted and has no observable beta of its own, and in any event a generation project is a different business.

Beta	What it measures	Where it comes from
Comparable company equity beta (also called the geared or levered beta)	Operating risk and financial risk of a listed comparable	Observed from the comparable's share price movements against the market
Underlying beta (also called the asset or ungeared beta)	Operating risk only, with the effect of gearing removed	Calculated by ungearing the comparable's equity beta at the comparable's capital structure
Debt beta	The systematic risk of the debt itself	Usually assumed to be nil. If the debt is genuinely risky, assuming nil understates the asset beta
Project equity beta (regeared)	Operating risk of the project plus the financial risk of the project's own capital structure	Calculated by regearing the asset beta at the project's target debt to equity mix

The formulas

- Ungear, assuming a debt beta of nil: $\beta_{\text{asset}} = \beta_{\text{equity}} \times E / (E + D(1 - t))$
- Regear: $\beta_{\text{equity}} = \beta_{\text{asset}} \times (1 + D(1 - t) / E)$
- Ungear where a debt beta is given: $\beta_{\text{asset}} = \beta_{\text{equity}} \times E / (E + D(1 - t)) + \beta_{\text{debt}} \times D(1 - t) / (E + D(1 - t))$
- Without tax: $\beta_{\text{asset}} = \beta_{\text{equity}} \times E / (E + D)$
- CAPM: $K_e = R_f + \beta_{\text{equity}} (R_m - R_f)$
- WACC = $K_e \times E / (D + E) + K_d (1 - t) \times D / (D + E)$
- Pre-tax equivalent, as an approximation only: $\text{pre-tax rate} = \text{post-tax WACC} / (1 - t)$

Worked example

Step	Calculation	Result
1. Comparable listed renewable energy producer	Equity beta 0.90, debt to equity 60:40, tax 27%	$\beta_{\text{equity}} = 0.90$
2. Ungear at the comparable's structure	$0.90 / (1 + 1.5 \times 0.73)$	$\beta_{\text{asset}} = 0.430$
3. Regear at the project's target mix of 50:50	$0.430 \times (1 + 1.0 \times 0.73)$	$\beta_{\text{equity}} = 0.743$
4. Cost of equity, risk-free rate 9.5% and market risk premium 6%	$9.5\% + 0.743 \times 6\%$	$K_e = 13.96\%$
5. Cost of debt after tax, pre-tax cost 12%	$12\% \times 0.73$	$K_d = 8.76\%$
6. Weighted average cost of capital	$(0.5 \times 13.96\%) + (0.5 \times 8.76\%)$	$\text{WACC} = 11.36\%$
7. Pre-tax equivalent, for use with pre-tax cash flows	$11.36\% / 0.73$	15.56%

The inputs above are illustrative. Use the figures given on the day.

What to watch

- Use several comparables where they are given: ungear each one at its own capital structure, average the asset betas, then regear once at the project mix. A single comparable imports that company's own peculiarities.

- The comparable must match the project, not Axion. For the generation project the comparable is a renewable energy producer, not a data centre operator.
- Regear at the project's target capital structure, not at its opening structure and not at the group's.
- Match the rate to the cash flows: real rate with real cash flows, nominal with nominal, and pre-tax with pre-tax. Document 4 uses pre-tax cash flows at initial feasibility.
- The risk-free rate and market risk premium must be South African if the cash flows are in rand.
- Do not add an arbitrary premium on top of a beta-derived rate for risks that are already in the cash flows. Project-specific risks belong in the cash flows or in sensitivity analysis, not in the discount rate.

Potential APC question 1 - Evaluate additional PPA versus direct investment

Answer framework

1. Purpose and context: state the electricity requirement, Axion 2030 link and decision objective.
2. Strategic fit: core competencies, management capacity, control and sustainability.
3. Financial implications: relevant lifetime cost, NPV, capital required and opportunity cost.
4. Risk allocation: compare construction, operating, generation, price, and volume risk.
5. Reliability: address intermittency, wheeling, residual grid supply and resilience infrastructure.
6. Flexibility: compare contractual lock-in with long-lived asset commitment.
7. Funding: assess gearing, liquidity, covenants and impact on data-centre growth.
8. Brief technical watchlist: accounting/tax only where material.
9. Recommendation: choose, justify and specify conditions/mitigations.

Conclusion template

ADAPTABLE CONCLUSION

On the available information, [OPTION] is preferable because [three Axion-specific reasons]. The conclusion is most sensitive to [key variables]. Approval should therefore be conditional on [due diligence / negotiated protections / funding and demand conditions].

Potential APC question 2 - Evaluate a specific PPA proposal

Answer framework: PRICE | VOLUME | TERM | SUPPLY | RISK | FLEXIBILITY

Heading	Points to apply to exam-day facts
PRICE	Base tariff; escalation; all-in delivered cost; wheeling/network costs; losses; residual grid cost; credits/unused power.
VOLUME	Forecast consumption; take-or-pay; pay-as-produced; excess production; actual utilisation ramp-up; time-of-use matching.
TERM	Duration; commencement; extension; repricing; alignment with Axion's facilities and strategy; long-term lock-in.
SUPPLY	Technology; output profile; availability; curtailment; grid connection; metering; non-delivery consequences.
RISK	IPP credit and completion risk; security; default; change in law; termination payments; guarantees.
FLEXIBILITY	Volume adjustments; additional sites; assignment; renegotiation; termination; ability to respond to changed demand.
DECISION	Compare against reference and direct-investment alternatives; identify break-even terms; recommend accept, reject or renegotiate.

High-value contract clauses to mention

- Tariff and escalation
- Volume commitment and unused electricity
- Commercial operation date and delay remedies
- Metering, loss and settlement methodology

- Curtailment
- Performance guarantees and non-delivery remedies
- Default, termination payments and security
- Change in law and regulatory cost allocation
- Environmental attributes / renewable certificates
- Assignment, extension and volume-flex rights

Potential APC question 3 - NPV / financial evaluation

Answer sequence

1. Identify the decision and ensure alternatives use the same time horizon and demand assumptions.
2. Identify relevant incremental cash flows and exclude sunk/common/non-cash items.
3. Use the appropriate project-specific discount rate supplied or derived as required i.e WACC.
4. Calculate NPV using a financial calculator. No NPV formula is included in this pack.
5. Interpret the result and compare alternatives on a consistent basis.
6. Perform sensitivity and break-even analysis on the variables that can change the recommendation.
7. Overlay capital affordability, funding, strategic fit, reliability and execution risk.
8. Conclude clearly and state conditions or safeguards.

Common calculation traps

Trap	Correction
Compare PPA tariff with direct Operating & Maintenance costs	Include direct CAPEX and all incremental lifetime costs.
Treat MW as annual energy	Use MWh/appropriate energy volumes for cash flows.
Assume every generated MWh is usable	Consider profile matching, losses, curtailment and excess electricity.
Ignore residual grid supply	Include the cost of supply required when renewables do not match load.
Ignore capital opportunity cost	Consider displaced data-centre investment and financial flexibility.
Accept on base-case NPV alone	Assess sensitivity, break-even thresholds and qualitative risks.

Interpretation template

ADAPTABLE CONCLUSION

The base-case NPV indicates [value created/destroyed] relative to [reference/alternative]. However, the result is most sensitive to [variables]. The option should proceed only if [break-even conditions and qualitative safeguards] are satisfied.